GROUP WORK PROJECT # 9284 MScFE 652: PORTFOLIO MANAGEMENT

GROUP NUMBER: __3_________

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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

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Project 3

Introduction

Improvements in Portfolio Optimization Using Denoising, Clustering, and Back-testing

Portfolio optimization strategies such as Markowitz Optimization, Black-Litterman and Kelly Criterion are well known but they suffer from some assumptions and may not perform well under noisy real data inputs, limited sample size vs. no of stocks or features and unstable correlation or covariance. We aim to improve upon our previous implementation by implementing denoising, clustering, and back-testing:

Improvement using Denoising

Denoising is the process of filtering out noise i.e. random data or inappropriate data fluctuations from the covariance matrix or returns. The most popular denoising technique is Random Matrix Theory (RMT) which separates signal form noise in the covariance matrix

- Improved Covariance Estimation:

Empirical Sample covariance matrix from original data can be unstable due to noise or esp. when no of assets is large than the no of observations. Fluctuations in one or two assets may dominate the whole equation. If the correlation between assets is high, then the total portfolio risk becomes concentrated. We can use PCA and keep only significant eigen values as those above the noise threshold predicted by RMT. Reconstruct a clean correlation matrix using only those components. Denoising stabilizes these estimates improving optimization of portfolio and allocation decisions

- Reduced Overfitting

Noisy data can lead to overfitting where the model performs well on past historical or train data but performs poorly on unseen test data or future data. Denoising mitigates this risk by retaining only the meaningful components i.e. eigen values that would represent the market dynamics close to reality.

- Better Risk adjusted Returns

By making use of denoised matrices, portfolios tend to have lower volatility and improved Sharpe ratio. This process is known as Shrinkage. This is important particularly for Kelly criterion where estimation errors can lead to overly aggressive positions.

Shrinkage means moving (or shrinking) an unreliable estimate toward a more reliable or structured target. The Empirical Sample Covariance Matrix is calculated purely from historical data. It can be very noisy, especially with short data histories or many assets.

A shrinkage estimator combines the SCM with a structured target, like the identity matrix or constant correlation matrix, in a weighted average.
$\Sigma shrinkage=(1-\lambda)\cdot\Sigma sample+\lambda\cdot\Sigma target$

where

- Σshrinkage is the improved covariance matrix.
- lambda [0,1] = **shrinkage intensity**.
- Σsample is the sample covariance matrix.

  $\Sigma sample=1/T-1*(R-Rm)^T(R-Rm)$

- R is matrix of returns and Rm is mean returns vector
- Σtarget is a structured matrix (e.g., identity, equal correlation).

In **Markowitz portfolio optimization**, the optimal weights are given by:

$w=\Sigma$^$-1\cdot(\mu-rf)$

where:

- w = vector of optimal portfolio weights
- Sigma^{-1} = **inverse of the covariance matrix**
- mu = expected returns
- rf = risk-free rate (downloaded along with data of assets)

Imagine trying to balance a fragile, cracked plate (sample covariance matrix) on your head. A small breeze (data error) knocks it off. But, a solid, smooth one (shrinked covariance matrix) stays steady.
The inverse of the covariance matrix is an important step in mean-variance optimization.
But **inverting** a noisy matrix leads to unstable, unreliable results**.** Shrinkage reduces noise and ensures that the inverse is stable**,** well-behaved and reflects true relationships between assets**.**

- Robust performance in high dimensions

In high dimensions cases like many assets and few observations, denoising allow optimization algorithms to function effectively.

1.2 Improvement using clustering

Clustering is the method used for grouping similar assets based on return profiles, sectoral data, or statistical similarity (e.g., correlation) or some closeness of distance metric (e.g. Euclidean). Examples are techniques such as Hierarchical Risk Parity (HRP), K-means, or Spectral Clustering:

- Improved Diversification
  Holding many assets doesn't guarantee true diversification. Traditional mean-variance optimization can result in portfolios that are not truly diversified in terms of risk exposure. Some assets may dominate over others. Clustering ensures assets with similar behavior are grouped, so capital is spread more evenly across distinct risk clusters.
- Reduction in Dimensionality
  Clustering can reduce the number of variables by working with clusters instead of individual assets, results in better optimization and reducing estimation error.
- Better Interpretability
  Clustered portfolios offer more interpretability to investors by showing which economic or sectoral themes the portfolio should be exposed to.
- Enhanced Robustness
  Clustering aids in constructing robust portfolios that are less sensitive to minor changes in the input data, especially useful in Black Litterman model where views can be applied at the cluster level.
- Use in Risk Parity Models
  Hierarchical clustering has been shown to outperform traditional methods in terms of stability and risk-adjusted returns. This can bring improvement upon strategies like Kelly Criterion and Markowitz.

Improvement using back-testing

Back-testing involves evaluating the performance of one or more strategy using historical data. It's an important step in assessing how the optimized portfolios would have performed under real market conditions.

- Validation of Model Assumptions
  By applying past data, we can verify whether the assumptions behind strategies like Black-Litterman or Kelly hold in practice and what impact does adjustments and constraints have on outputs.
- Strategy Comparison

Back testing allows side-by-side evaluation of different models (e.g., Markowitz vs. Black-Litterman vs. Denoised+Clustered models) using consistent performance metrics such as Sharpe ratio, drawdown, CAGR, and volatility.

- Identification of Overfitting:
  A strategy that performs well in-sample but poorly out-of-sample is likely overfitted. Back testing over multiple rolling windows helps identify and correct such issues.
- Improved Parameter Selection:
  Optimization models require parameters (e.g., expected returns, views, confidence levels). Back testing helps in tuning these parameters by assessing which combinations yield the best performance.
- Stress Testing and Scenario Analysis:
  Back testing can include simulations through historical crises (e.g., 2007-08 meltdown, COVID crash) to see how strategies behave in stressed markets. This is particularly important for aggressive strategies like Kelly Criterion.
- Realistic Performance Estimation:
  Including realistic transaction costs, slippage, and rebalancing frequency during backtesting offers a more accurate picture of real-world performance and helps adjust the model accordingly.

Methodology

Steps for Denoising (RMT)

- Compute Correlation Matrix
- Apply Eigen Decomposition using correlation matrix
- Identify noise threshold (lambda) using Marchenko Pastur limit
- λ_=(1+1/sq_rt(Q))^2where Q=T/N
- Filter out noisy eigen values
- Reconstruct Denoised correlation matrix
- Convert to denoised covariance matrix
- Use cov_denoised in MVO, Kelly

Steps for Clustering

- Compute Correlation matrix and distance matrix

- Convert to condense distance
- Perform hierarchical clustering
- Form flat clusters
- Map cluster labels to stock tickers
- Use the weights of the assets in the portfolio as the risk parity.

Denoise + Clustering

We can use both to get refined input for MVO or Kelly models. We used the two types of the denoised covariance matrix to construct a portfolio using MVO and Kelly. Also we used these two denoised correlation as inputs to the HRP clustering. The idea is to have a more robust estimation of the correlation compared to the sample data.
We also used the result of the clustering to select a representative of each cluster and construct a portfolio using them.

Best Previous Results

We use the data from 2023-01-01 to 2025-04-30 to train our models and testing data is from 2025–05-01 to 2025-06-01.

Our portfolio is composed of the following firms and assets. The firms are selected from each sector of S&P 500. The Gold and Bitcoin assets were included to have an almost complete universe of options. We avoided the assets from other countries to simplify the process.

We also don't consider the short sell positions.

Sectors/company					
Materials	LIN (Linde plc)	APD (Air Products and Chemicals Inc.)	ALB (Albemarle Corporation)		
Communication Services	META (Meta Platforms Inc.)	CMCSA (Comcast Corporation)	FOXA (Fox Corporation)	GOOGL (Alphabet Inc.)	
Energy	XOM (Exxon Mobil)	COP (ConocoPhillips)	APA (APA Corporation)	CVX (Chevron Corporation)	

Financials	JPM (JPMorgan Chase & Co)	PNC (PNC Financial Services)	LNC (Lincoln National)	BRK-B (Berkshire Hathaway)	V (Visa Inc.)
Industrials	CAT (Caterpillar Inc.)	UPS (United Parcel Service)	CAPR (Carrier Global Corporation)	GE (General Electric)	
Information Technology	NVDA (NVIDIA Corporation)	SNPS (Synopsys Inc.)	ZS (Zscaler Inc.)	AAPL (Apple Inc.)	MSFT (Microsoft)
Consumer Staples	PG (Procter & Gamble Co.)	KMB (Kimberly-Clar k Corp.)	TAP (Molson Coors Beverage)	COST (Costco Wholesale Corp.)	
Utilities	NEE (NextEra Energy Inc.)	DUK (Duke Energy Corp.)	AES (AES Corp.)		
Health Care	UNH (UnitedHealth Group)	MDT (Medtronic plc)	BAX (Baxter International)	LLY (Eli Lilly)	
Consumer Discretionary	AMZN (Amazon Inc)	NKE (NIKE Inc.)	ETSY (Etsy Inc.)	TSLA (Tesla Inc.)	
Real Estate	PLD (Prologis Inc.)	AMT (American Tower Corp)	AVB (AvalonBay Communities)		
Extra Assets	GC=F (Gold)	BTC-USD (Bitcoin)			

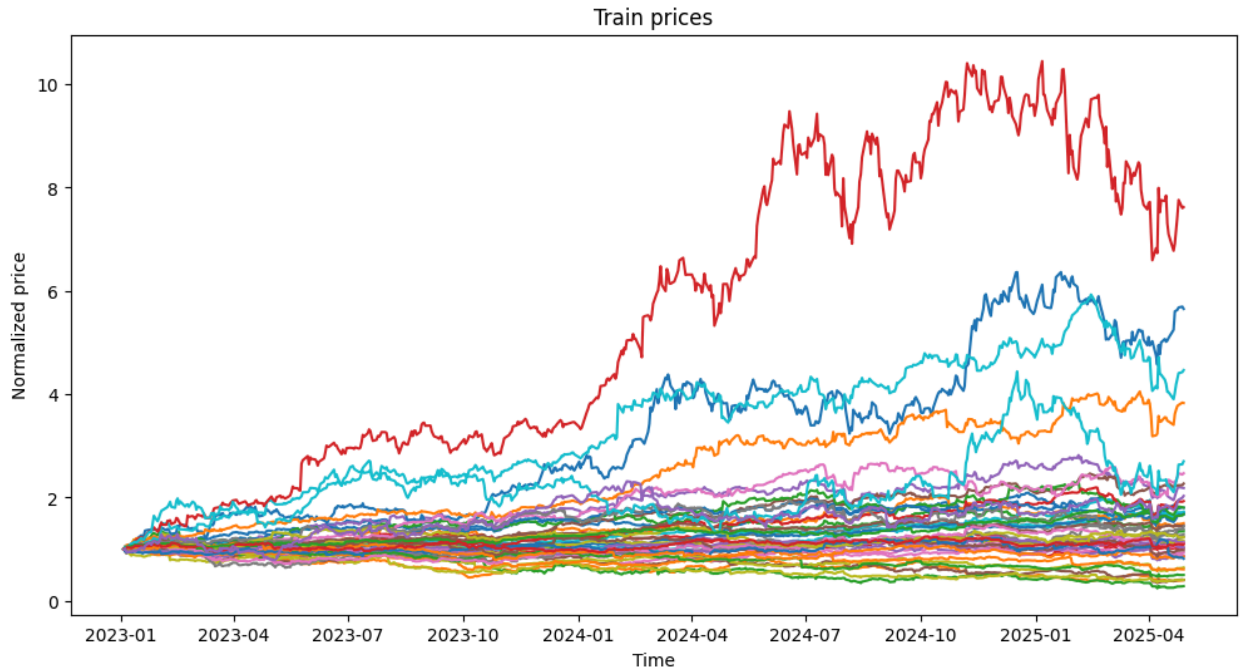
- We used the Close feature column and calculate the geometric returns as formula:

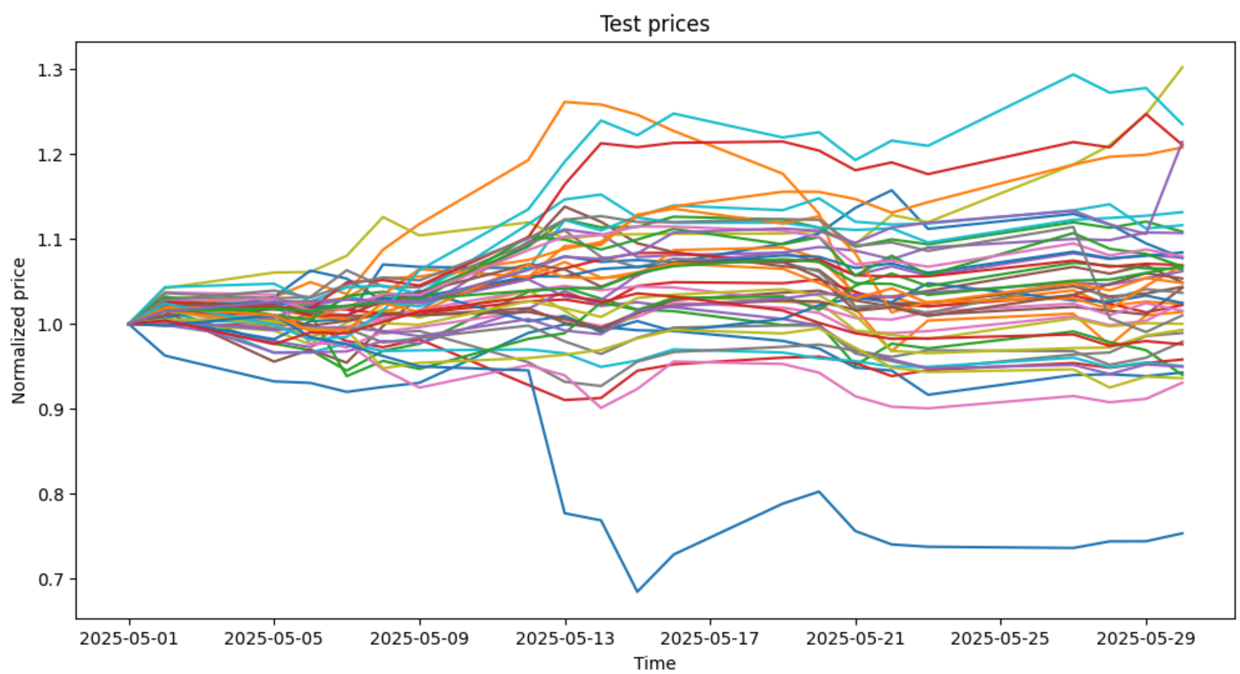
  $r_t = \log(P_t/P_{t-1})$

  then we calculated mean returns.

- We downloaded daily 2-year Treasury yield from the FRED database (DGS2 series) as Risk Free Rate of return. Converted annual rate to daily risk free rate by dividing by trading days (as 252). Then we calculated the mean daily risk free rate.

Normalized Stock prices growth Plots for Train and Test data:

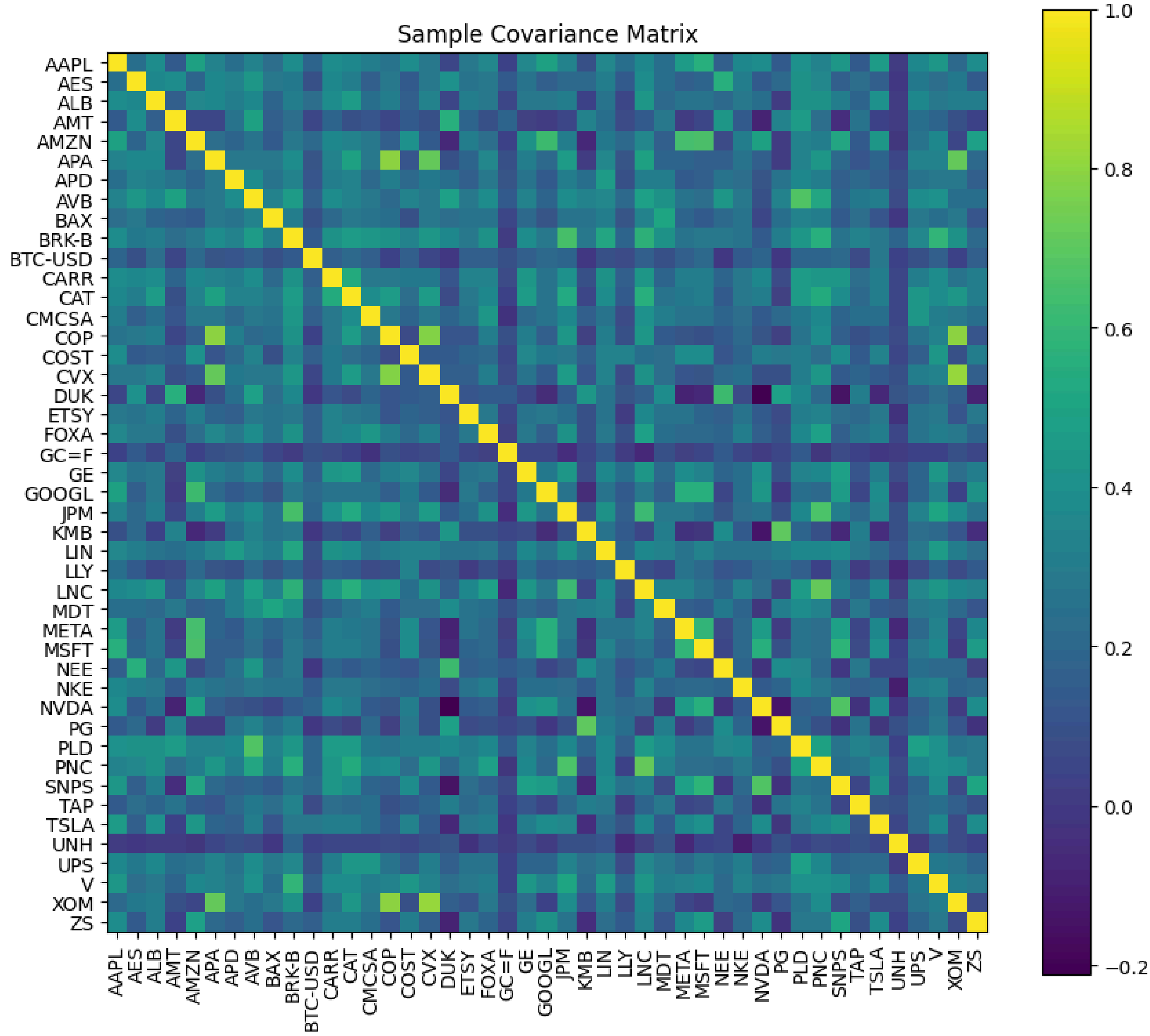




In the GWP2 our best result was using the Kelly Full approach. However, to have a complete comparison here we will also use the mean-variance (Markowitz) optimization (MVO).

Sample Covariance

Using the historical data we get the following covariance matrix

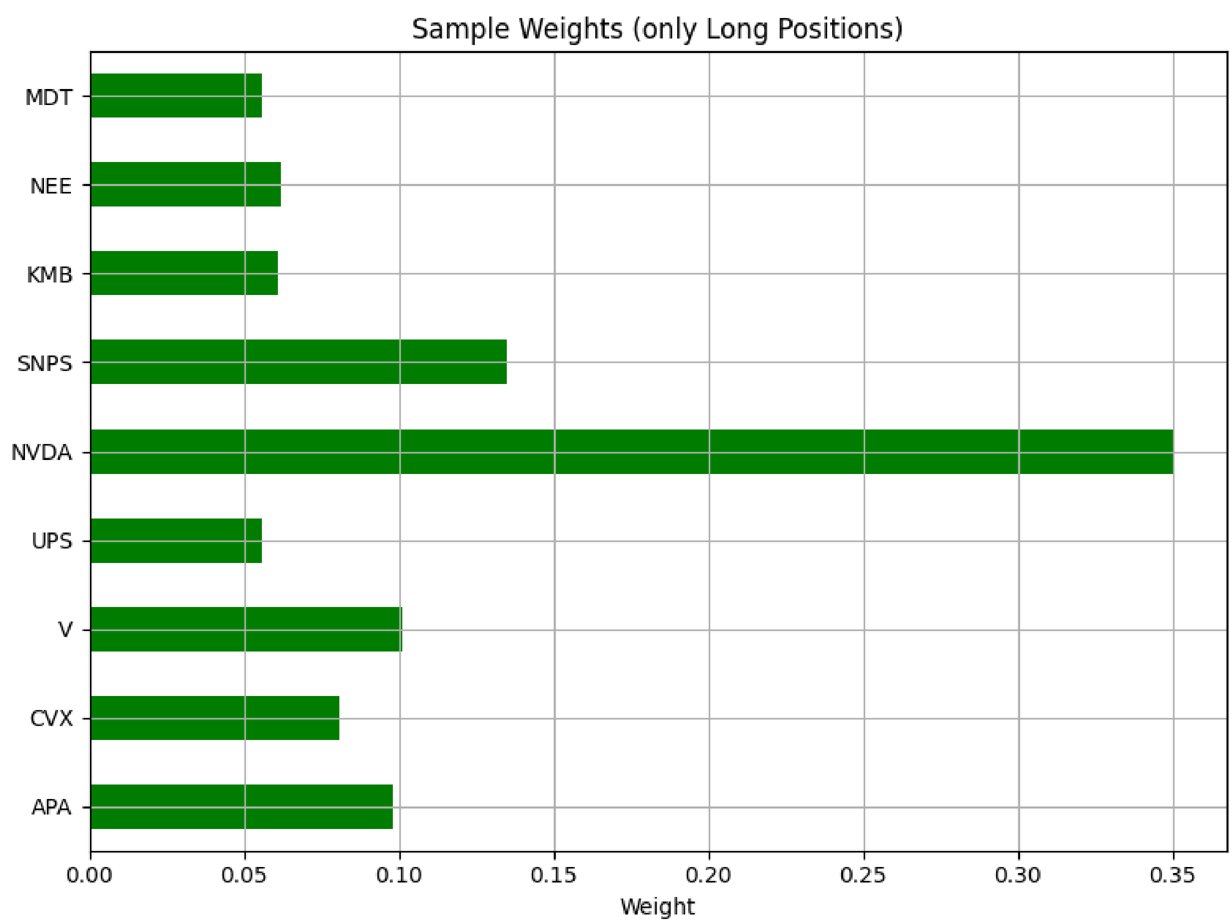


MVO optimization

After using the MVO optimization we get the following non-zero weights

	MVO (Sample)
APA	0.098
CVX	0.081
V	0.101
UPS	0.056
NVDA	0.350
SNPS	0.135
KMB	0.061
NEE	0.062
MDT	0.056

Or graphically



The result of this portfolio during the test period is

```
Portfolio stats:
```

	SampleΣ	
	Train	**Test**
μ_ann	0.254	1.138
σ_ann	0.254	0.242
Sharpe	1.000	4.710
TotalReturn	0.845	0.095

Full Kelly Optimization

The Kelly optimization gave us the following weights which indicate that the portfolio is't diversified at all. In this report, we have bound the maximum weight of each asset to 0.35 to reduce the greedy behaviour of Kelly approach and construct more diversified portfolios.

	Kelly (Sample)
Ticker	
BTC-USD	0.35
NVDA	0.35
META	0.30

The result of this portfolio during the test period is

```
Portfolio stats:
```

	Kelly	
	Train	**Test**
μ_ann	0.722	1.602
σ_ann	0.347	0.258
Sharpe	2.081	6.211
TotalReturn	4.085	0.136

Clearly this result is better than the portfolio constructed by MVO. Here we should be aware of the high risk of the portfolio. Later, comparing this portfolio with other ones we will see this point better.

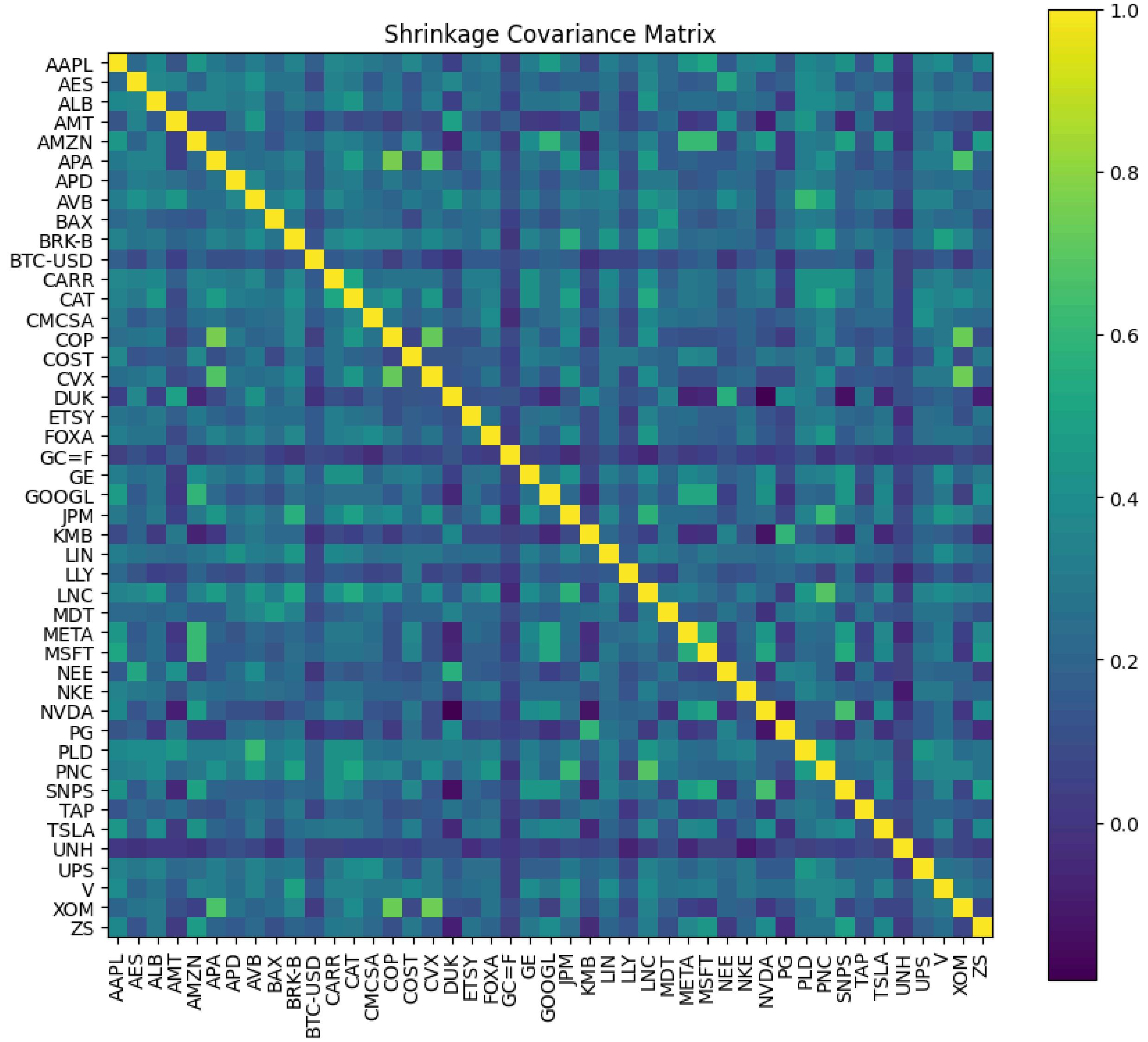
Denoising Improvements

In this section we employ some denoising methods to get a more robust estimation of the covariance matrix.

Ledoit–Wolf Shrinkage

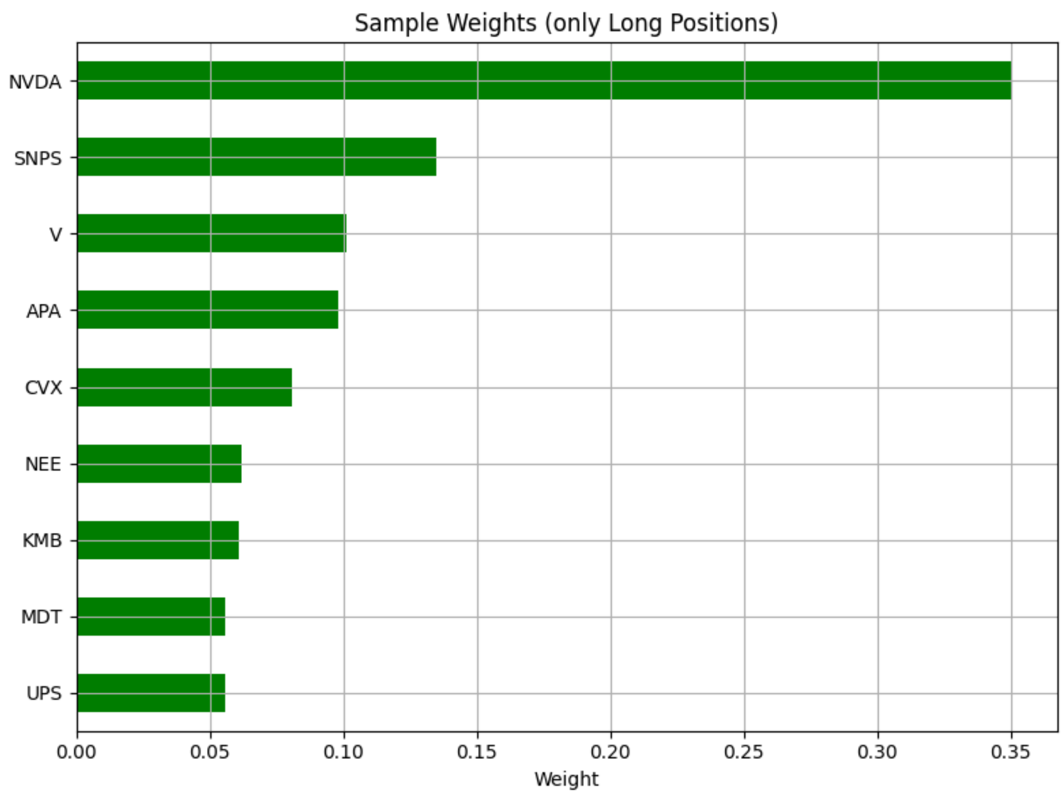
Shrinkage does reduce the noisy correlation and covariance so we implemented Ledoit–Wolf as our shrinkage estimator. We will compare the shrinkage covariance results from the MVO with the sample empirical covariance results.
We are trying to maximize the Sharpe ratio on returns of assets as our optimization equation. One constraint that the sum of normalized weights should be equal to one.

The Shrinkage covariance matrix is as follow



Below is the MVO output from shrinkage covariance matrix

	MVO (Shrinkage)
APA	0.081
CVX	0.087
V	0.096
UPS	0.049
NVDA	0.350
SNPS	0.141
KMB	0.065
NEE	0.069
MDT	0.061



The result of this portfolio during the test period is

```
Portfolio stats:
```

	MVO Shrinkage	
	Train	Test
μ_ann	0.261	1.132
σ_ann	0.252	0.239
Sharpe	1.037	4.736
TotalReturn	0.879	0.095

If we use the Kelly optimization we get the following weights

	Kelly (Shrinkage)
Ticker	
BTC-USD	0.35
NVDA	0.35
META	0.30

Which results in the following statistics

```
Portfolio stats:
```

	Kelly Long Only Fully Invested Shrinkage	
	Train	Test
μ_ann	0.722	1.602
σ_ann	0.347	0.258
Sharpe	2.081	6.211
TotalReturn	4.085	0.136

Denoising Correlation by RMT

Another denoising method is using the random matrix theory to remove the eigenvalues related to the noisy signal and detect the real ones.

The weights of the MVO optimization for a denoised correlation matrix is as follow

	MVO RMT
NVDA	0.114
APA	0.091
V	0.089
SNPS	0.074
AVB	0.065
AAPL	0.050
NEE	0.049
KMB	0.043
PG	0.042
GE	0.038
DUK	0.037
MDT	0.035
CMCSA	0.034
LIN	0.033
CVX	0.033
ZS	0.028
XOM	0.026
UPS	0.025
JPM	0.017
BTC-USD	0.014
TSLA	0.012
PNC	0.012
NKE	0.012
ETSY	0.010
COST	0.006
BAX	0.005
TAP	0.004

As it can be seen clearly, in this approach we have a more diversified portfolio. These weights result in a out-of-sample performance as below

```
Portfolio stats:
```

	MVO RMT	
	Train	Test
μ_ann	0.290	0.630
σ_ann	0.146	0.118
Sharpe	1.985	5.348
TotalReturn	1.106	0.054

The Kelly optimization for this denoised covariance matrix gives us the following results

```
Ticker
BTC-USD    0.35
NVDA       0.35
META       0.30
dtype: float64
```

Which performs as

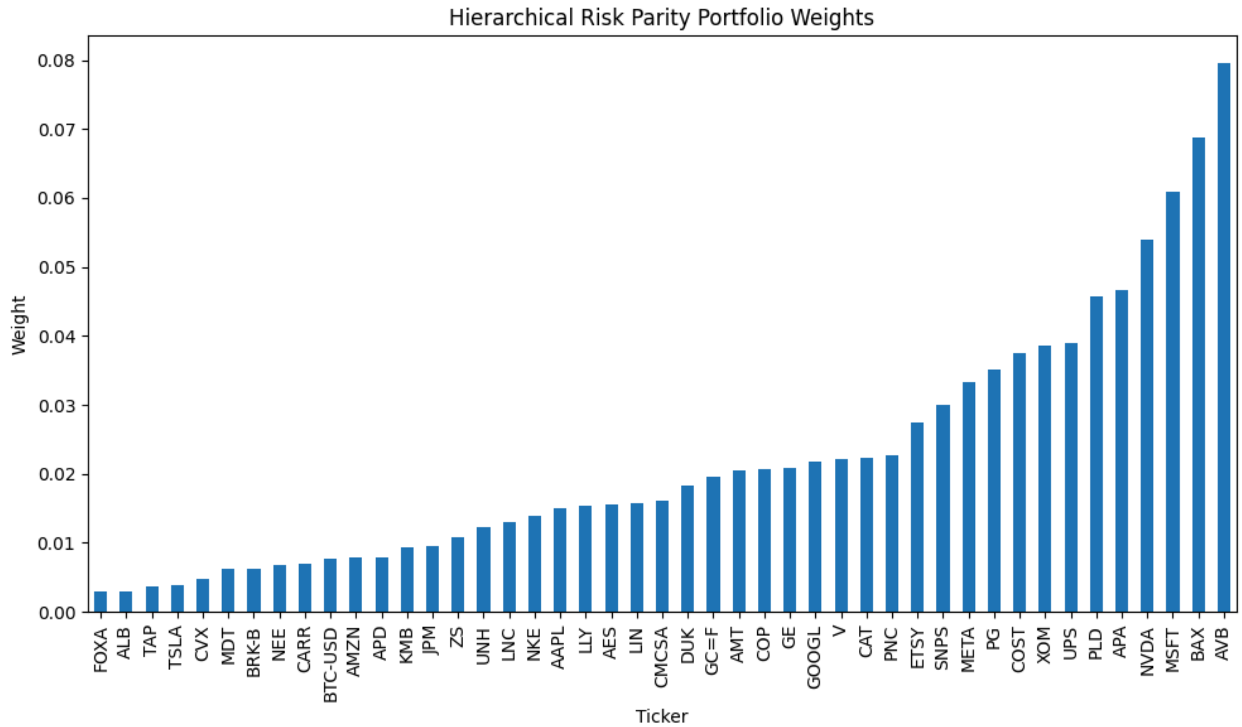
```
Portfolio stats:
```

	Kelly RMT	
	Train	Test
μ_ann	0.722	1.602
σ_ann	0.347	0.258
Sharpe	2.081	6.211
TotalReturn	4.085	0.136

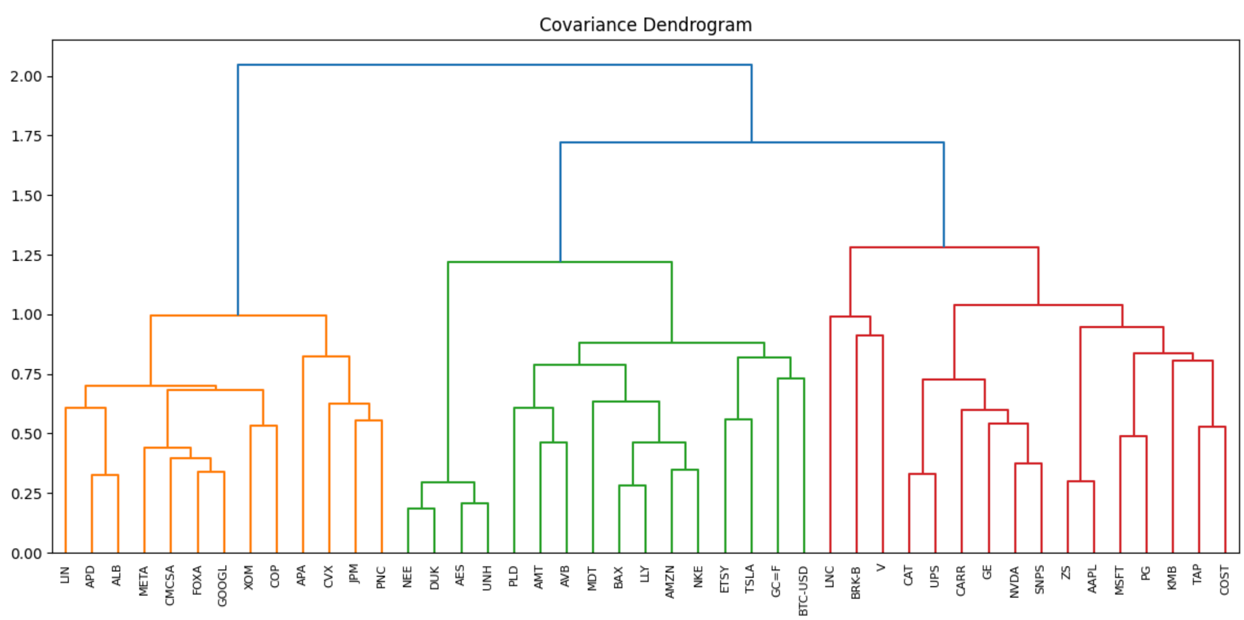
Clustering Improvements

Hierarchical Risk Parity (HRP)

First we use the sample correlation matrix to apply the **HRP** clustering and choose the weight of the assets in the portfolio. The graphical result is as follow



To have a better understanding of the clusters it's worthwhile to take a look at the dendrogram plot of the clusters.

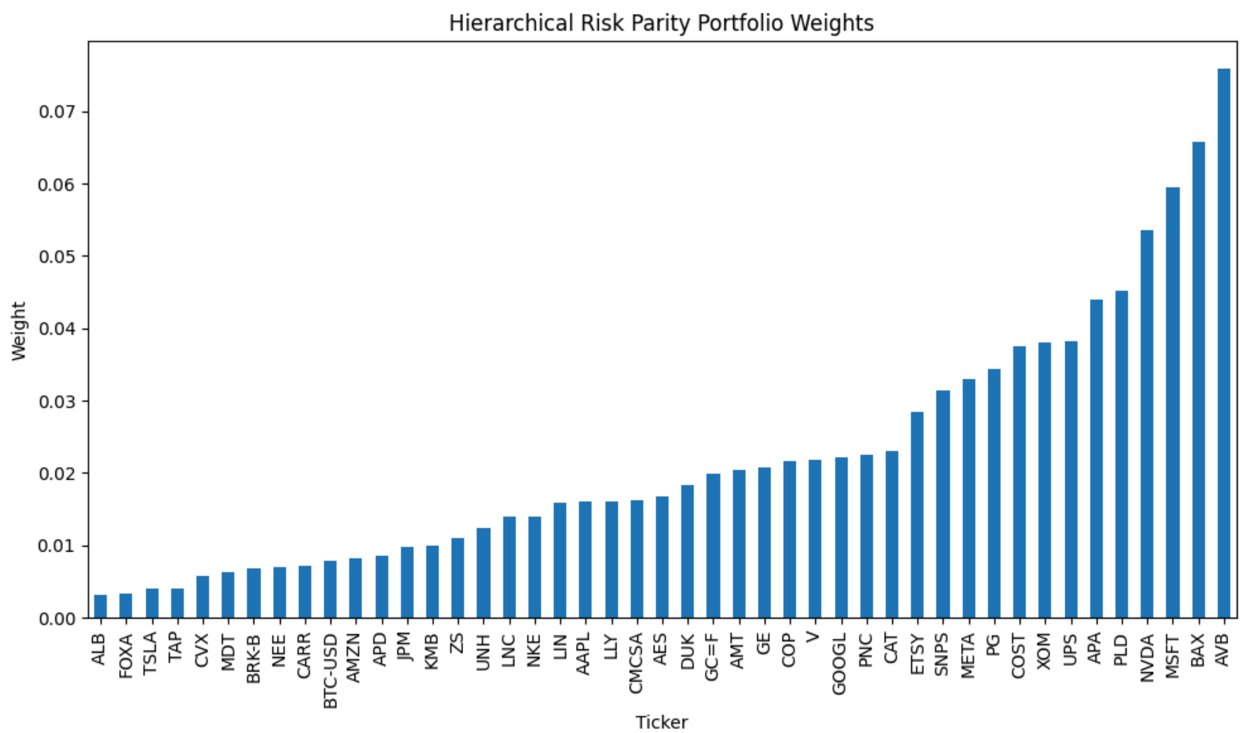


The performance of this portfolio is as follow.

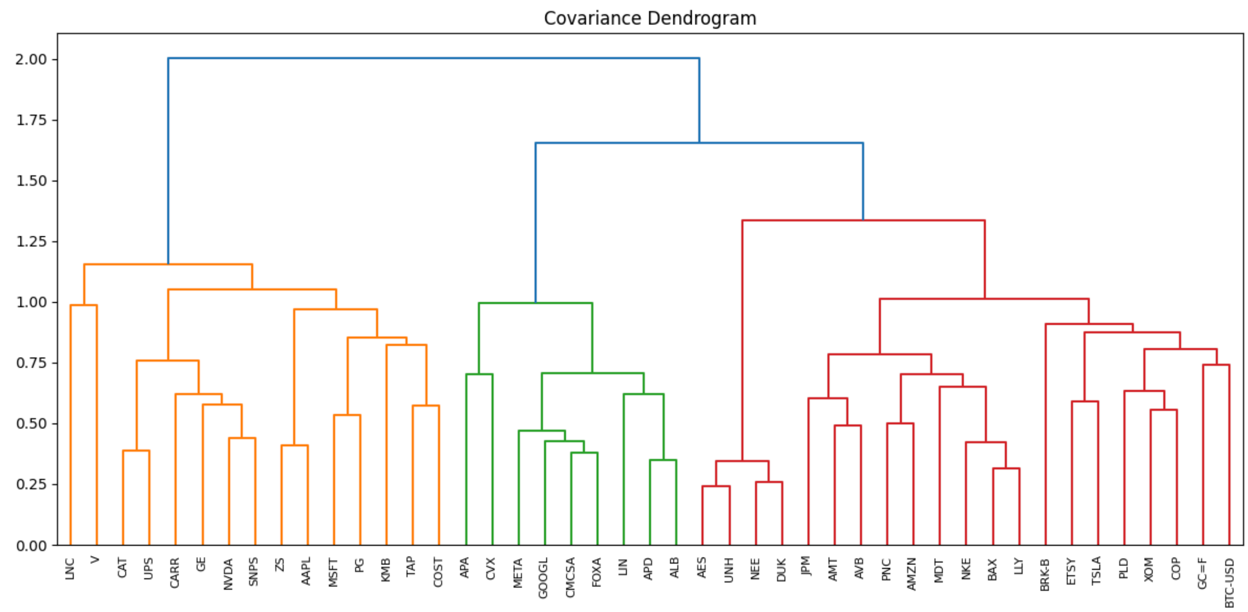
```
Portfolio stats:
```

	HRP	
	Train	**Test**
μ_ann	0.135	0.636
σ_ann	0.157	0.159
Sharpe	0.863	4.004
TotalReturn	0.470	0.054

The interesting point about the above result is the very low volatility of the portfolio, as expected from a risk parity approach. If we use the shrinkage covariance matrix we get the following weights



The dendrogram plot is

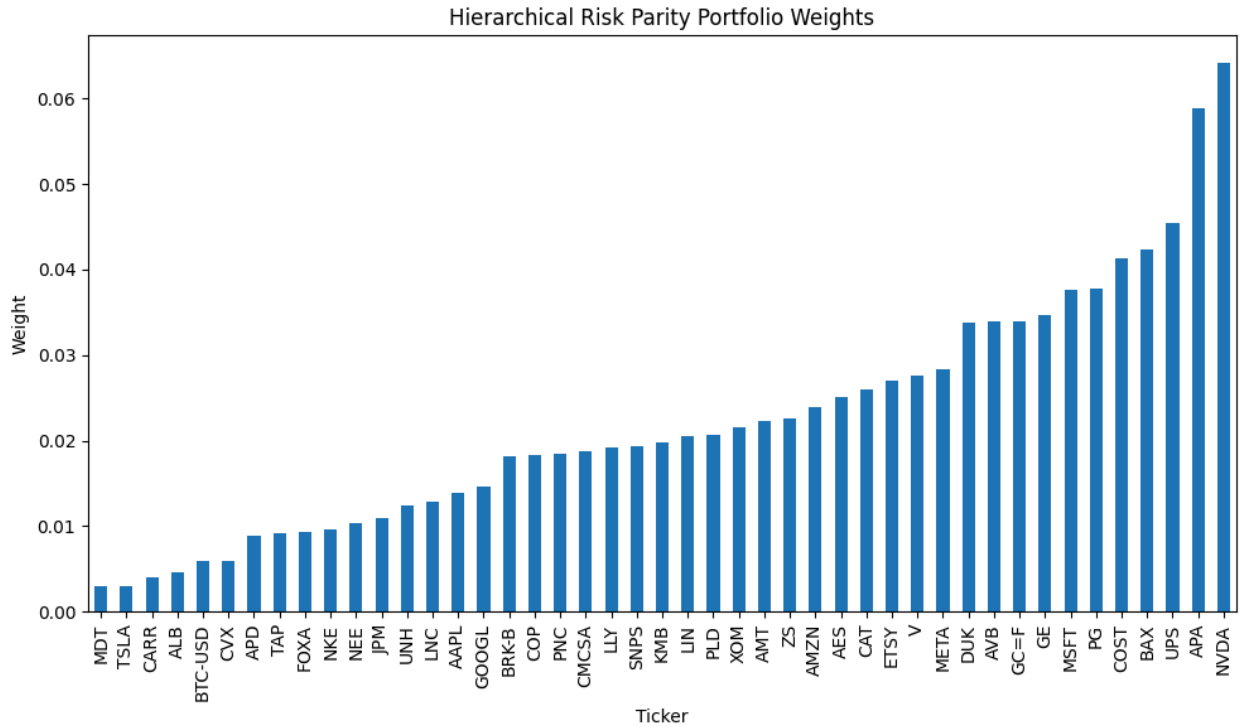


The performance of this portfolio can be summarized as

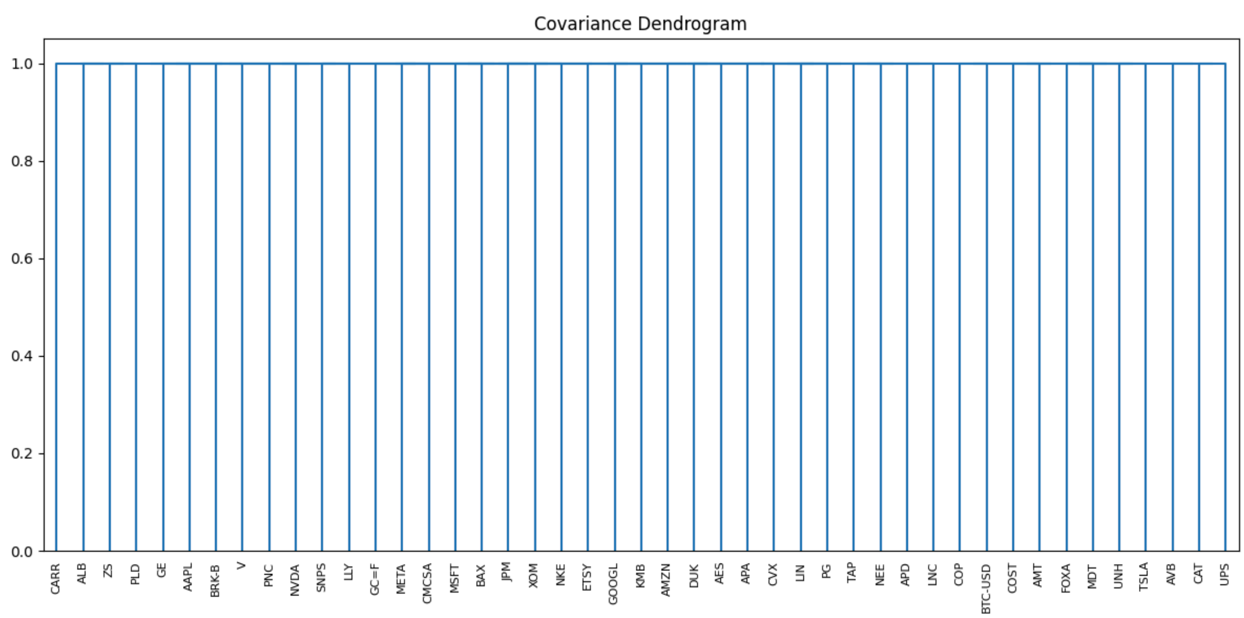
Portfolio stats:

	HRP	
	Train	Test
μ_ann	0.135	0.630
σ_ann	0.156	0.159
Sharpe	0.861	3.965
TotalReturn	0.468	0.054

Another interesting approach is using the denoised covariance by RMT to do the clustering.



To plot the dendrogram of this clustering we had to symmetrize the denoised correlation matrix. The result was not what we expected. It seems that there are some subtleties in computing the dendrogram. We think that the below plot isn't informative but, for sake of completeness we have added it to the report.



We got the performance of the portfolio as below

```
Portfolio stats:
```

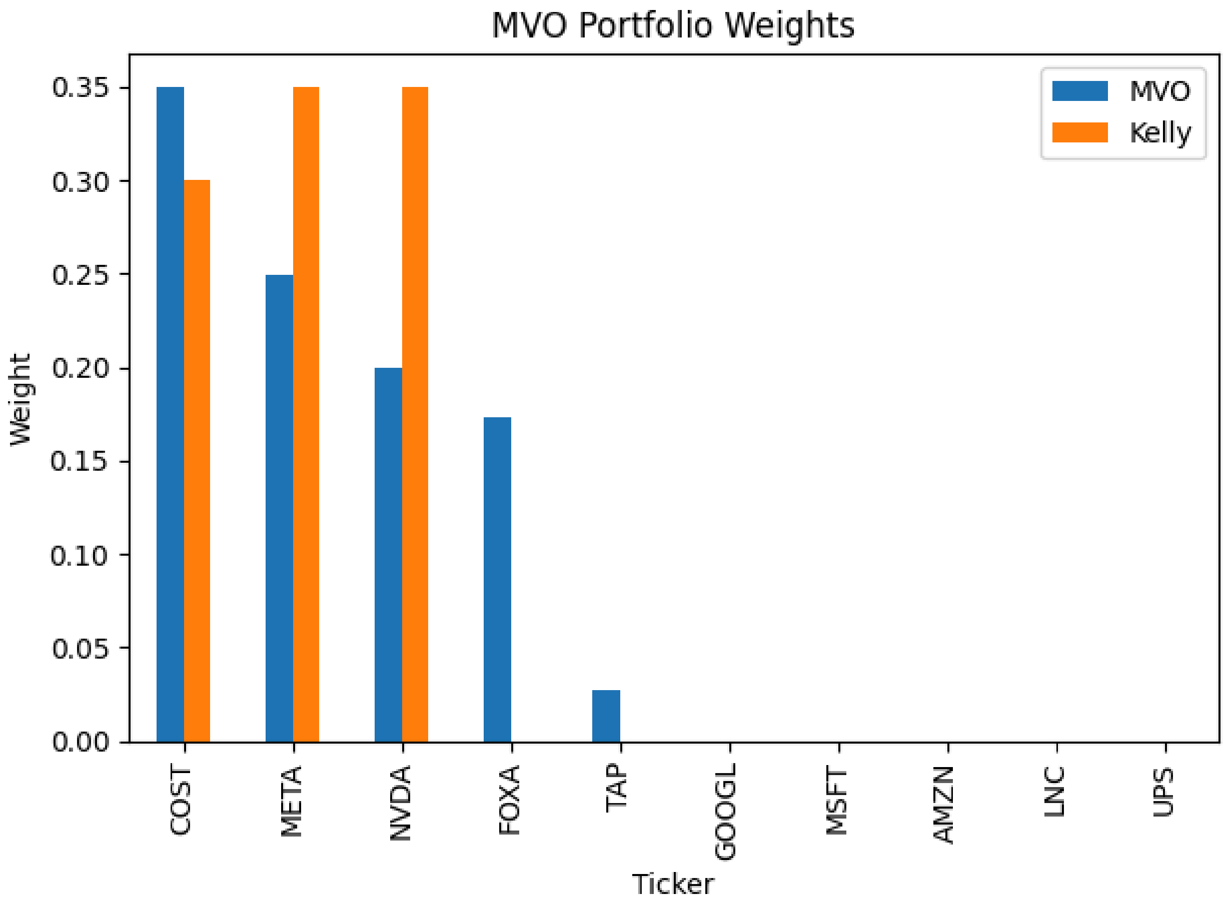
	HRP	
	Train	Test
μ_ann	0.016	0.362
σ_ann	0.158	0.164
Sharpe	0.102	2.212
TotalReturn	0.117	0.031

Representative selection of clustering

Another possibility is to do the clustering and select one representative of each cluster. Here we tested several configurations by changing the method of linkage and also the number of clusters. For the ward linkage and 10 clusters we got the following representative assets to input into our optimization procedures

```
Cluster representatives: ['META', 'TAP', 'GOOGL', 'UPS', 'COST', 'LNC',
'FOXA', 'AMZN', 'NVDA', 'MSFT']
```

```
Using the MVO and Kelly optimizations for these assets we arrived at the
following weights
```



The performance of these portfolios are

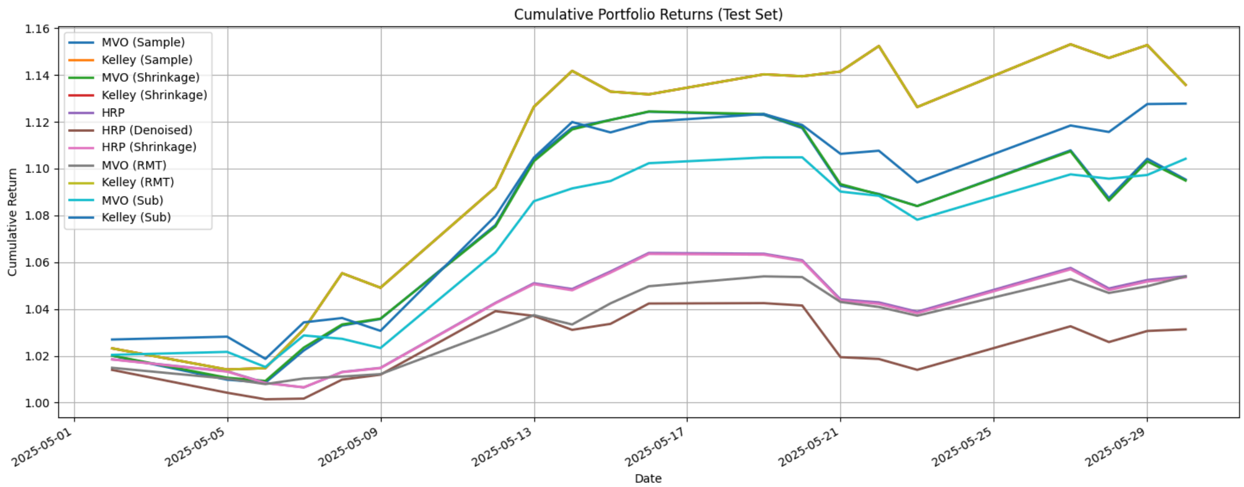
	MVO
μ_ann	1.230651
σ_ann	0.195512
Sharpe	6.294507
TotalReturn	0.104203

	Kelly
μ_ann	1.506525
σ_ann	0.237710
Sharpe	6.337654
TotalReturn	0.127755

The Kelly optimization were highly greedy to select the assets and sacrificed the volatility to maximize the return.

Comparing the Portfolios Performance

The cumulative portfolio returns can be seen in the below figure

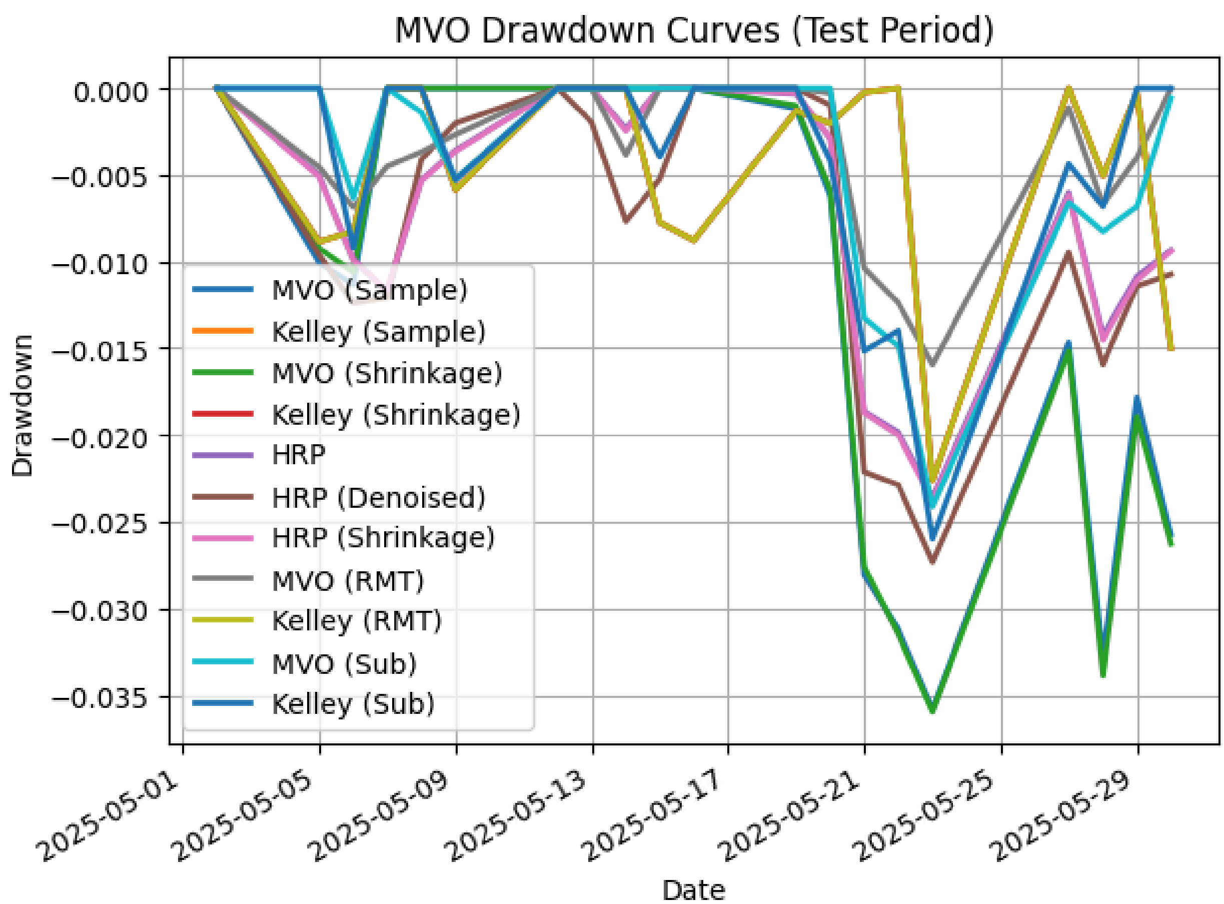


One important statistics is to verify the maximum drawdown of each portfolio

📉 Maximum Drawdowns (%):

	Max Drawdown (%)
MVO (Shrinkage)	-3.60
MVO (Sample)	-3.58
HRP (Denoised)	-2.73
Kelly (Subset)	-2.60
MVO (Subset)	-2.41
HRP (Shrinkage)	-2.38
HRP	-2.36
Kelly (Sample)	-2.26
Kelly (RMT)	-2.26
Kelly (Shrinkage)	-2.26
MVO (RMT)	-1.59

These drawdowns during the testing period are as follow



Summarising the out-of-sample performance statistics on of all portfolios is

	Ann. Return	Ann. Vol	Sharpe	Total Ret	Max Drawdown (%)
Kelly (Sample)	1.602	0.258	6.211	0.136	-2.262
Kelly (RMT Denoised)	1.602	0.258	6.211	0.136	-2.262
Kelly (Shrinkage)	1.602	0.258	6.211	0.136	-2.262
Kelly (Sub)	1.507	0.238	6.338	0.128	-2.600
MVO (Sub)	1.231	0.196	6.295	0.104	-2.414
MVO (Sample)	1.138	0.242	4.710	0.095	-3.583
MVO (Shrinkage)	1.132	0.239	4.736	0.095	-3.596
HRP (Sample)	0.636	0.159	4.004	0.054	-2.361
HRP (Shrinkage)	0.630	0.159	3.965	0.054	-2.378
MVO (RMT Denoised)	0.630	0.118	5.348	0.054	-1.595
HRP (RMT Denoised)	0.362	0.164	2.212	0.031	-2.733

Conclusion and Final Remarks

Key Points:

- Kelly portfolios, overally, have the highest Sharpe ratios and total returns, with moderate to high drawdowns.
- MVO (Sub) has high Sharpe with relatively low drawdown and volatility.
- MVO (Sample) and MVO (Shrinkage) are nearly identical, showing high return, but more drawdown and volatility compared to the clustered or denoised versions. We had seen this pattern in the GWP2 too.
- HRP portfolios, as expected, have lower risk/lower return, but have very consistent and stable drawdowns.
- RMT denoising reduces the volatility and maximum drawdown in MVO/HRP, but can also reduce returns. Generally, it depends on the asset universe and market regime.
- HRP (RMT Denoised) is the most conservative, with the lowest return and Sharpe.

What reasons might explain these differences in performances?

Financial assets and Historical data

- Our asset universe has a **mix of equities types** (diverse sectors), gold, and Bitcoin. In the recent sample window, assets like NVDA and BTC-USD have extremely high returns, dominating Kelly/MVO portfolios.
- Clustered representatives (MVO/Kelly Sub) **were not well diversified**. The performance of these models probably can not be optimized in the long run.
- Our train period (Jan 2023–Apr 2025) clearly has a strong uptrend in tech/crypto bias models toward these.
- Our testing period (May–June 2025) is a **very short window**. Methods that chased the prior winners (Kelly, MVO) outperformed, but this is fragile and isn't reliable.
- **High correlations** within sectors make simple sample covariance **unstable**, as expected, favoring robust estimators.

Model structure and dynamics

- Sample Covariance (MVO, Kelly) are highly sensitive to estimation noise and recent extreme returns. It tends to concentrate on top performers if allowed.
- Shrinkage: Ledoit-Wolf shrinkage stabilizes the covariance matrix, reducing estimation noise, but has little effect on the result because extreme returns in training and testing data still dominate the mean vector.
- RMT Denoising filters out noisy eigenvalues, resulting in a lower effective dimensionality of the risk model. This stabilizes portfolio volatility/drawdown, sometimes at the expense of reducing the mean return if strong factors are removed. It is a good technique but must be calibrated carefully.
- Clustering + Representatives (Sub) dramatically reduces asset universe dimensionality and correlation redundancy. In our analysis, this boosted the Sharpe ratio, lowers volatility, and moderately increases return, because it keeps one "winner" per cluster and avoids over-concentration.
- HRP diversifies risk by construction, using correlation clusters. These portfolios are less sensitive to mean/cov noise, but tend to underperform pure MVO/Kelly if there's a strong trend regime. But, they are safer in a more volatile market.
- HRP with Denoising reduces risk further, but at the cost of lower return and Sharpe, as expected.

In which cases, if any, does the incremental gain in performance justify the additional complexity and effort?

When Complexity is Justified:

- RMT Denoising is justified if our portfolio has many correlated assets (e.g. from the same sector), high estimation risk, or we want to avoid large drawdowns. In our results, it reduces max drawdown (for MVO: from -3.58% to -1.60%) and volatility, but may also reduce return.
- Clustering and using representatives is strongly justified when you want to balance diversity and performance.

- The “Sub” portfolios (MVO/Kelly on cluster reps) give the highest Sharpe and return with lower volatility than the raw portfolios, and only slightly higher drawdown than denoised HRP/MVO.

- HRP approach is justified when an investor prioritizes stability and drawdown protection instead of chasing the highest returns, or when means/covariances are too unstable to trust MVO/Kelly at all.

When Complexity is Not Justified:

- Simple MVO/Kelly is sufficient if the investor has strong return dispersion and can tolerate estimation risk and occasional high drawdown moments.If the asset universe is small and not very correlated, denoising and clustering may be overkill and totally unnecessary.